

Bryant Hayden

Controls Engineer

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Location: Lebanon, OR

Summary

Full stack application developer with a B.S. in Computer Science and nearly two years building and supporting production SCADA, MES, and IIoT applications in a high-volume manufacturing plant. Works the full delivery cycle: validating requirements with operations stakeholders, writing specifications, developing in Ignition Perspective with Python/Jython and SQL, testing against documented plans on a dedicated development gateway, then commissioning, documenting, and training end users. Comfortable sitting between IT and Engineering and translating between them. Work developed at Stayton has set the integration precedent adopted across 31+ plants, with two production systems poised for rollout to 11 lines across 5 plants.

Technical Skills

- **Languages:** Python, Jython, SQL, JavaScript, Java, C, C++, C#, HTML/CSS
- **Industrial Software:** Ignition 8.1 (Perspective, Edge), SCADA, HMI, MES, UDTs, alarming, historian, OEE
- **Protocols & Integration:** MQTT, OPC UA, TCP/IP, REST APIs, Node-RED, Allen-Bradley PLCs, ERP to MES data flows, Revolution Pi edge gateways
- **Databases:** SQL Server, MySQL, schema design, query optimization, data pipelines and normalization
- **Tools & Practice:** Git / GitHub, Agile & Scrum boards, documented test plans, development vs. production environments, Visual Studio / VS Code, Linux, Windows Server
- **Other:** Cisco switching, VLAN segmentation, industrial networking, ISA-95 alignment, Industry 4.0

Professional Experience

IT Lead - Plant Applications & Automation

JELD-WEN | Stayton, OR | Nov 2024 - Present

- Sole developer and system owner for the Stayton Ignition SCADA environment, supporting a plant running 9 production lines at roughly 2,300 units per day
- Develop, test, and maintain Ignition 8.1 Perspective applications used daily in manufacturing operations, covering navigation, role-based security, alarming and diagnostic views, and reusable components
- Validate system requirements with production, maintenance, and CI stakeholders, then write the programming specifications that drive development
- Write backend logic in Python/Jython and SQL integrating ERP, MES, PLCs, databases, edge gateways, and vendor equipment over MQTT, OPC UA, TCP/IP, and REST APIs
- Develop and test on a dedicated development gateway separate from production, using documented test plans and version control to validate changes before release
- Collaborate on a cross-site development team, coordinating with corporate IT, controls engineering, vendors, and plant operations; author runbooks, SOPs, and architecture diagrams so systems can be supported by others
- Train end users from first-time computer users to senior stakeholders, driving adoption and measurably reducing support escalations
- Automated reporting with SQL, Python, and VBA, eliminating 12 daily manual reports and a recurring biweekly report to recover roughly 550 labor hours per year

Key Projects

SUDS — Soap Usage & Dispensing System (in production, multi-plant rollout planned)

- Built an end-to-end water quality monitoring and automated dosing system for the plant's glass washer, owning the stack from sensor to user interface

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- Delivered the 4-tank system for \$3,051 against a \$35,850 commercial vendor quote, a 91% cost reduction, with Ignition-native trending, alarming, and dosing control included rather than quoted separately
 - Diagnosed cross-sensor interference in the prototype and specified 4 industrial isolated conductivity probes and transducers for stable independent per-tank readings, moving the system from proof of concept to production grade
 - Developed the Python edge software and data normalization layer on a Revolution Pi gateway behind an Ignition Perspective front end for live monitoring, alarming, and remote control

Ignition Laser Etcher & Serialization System

- Developed a full stack Ignition and Python replacement for a deprecated Java etching platform, restoring serialized traceability and projected at \$143K+ savings over 3 years, architected toward a 5-site, 11-line rollout
- Built an operator interface handling batches of roughly 5,000 records across 500 pages with responsive navigation, supporting lines etching thousands of lites per day
- Gathered requirements with operations and quality, then built across PLC tag integration, Jython scripting, SQL-backed configuration, vendor equipment communications, and ERP to MES data flow

Plant Floor Visibility Platform

- Designed a real-time Ignition Perspective floor map and dashboard suite surfacing live equipment status, downtime conditions, alarms, and KPIs in a single interface
- Integrated OPC UA endpoints alongside MES data; presented the platform to senior leadership and to the full North America plant manager network as a designated best practice, driving inbound adoption requests from other sites

MES (L2L) Implementation & API Integration

- Led site rollout of L2L production, downtime, quality, and reporting modules aligned to enterprise standards
- Developed against the externally hosted Leading2Lean REST API to move production data programmatically, with data mappings that automated machine data capture and eliminated manual operator entry

IIoT Edge Gateway Standardization

- Standardized Revolution Pi deployments using reusable images and Python / Node-RED configurations, bringing legacy equipment into MQTT and OPC UA data streams
- Reduced per-unit deployment to roughly 2 hours through a documented 3-step playbook, now the model for L2L machine integration across 31+ plants in North America

Additional Technical Projects

Autonomous Rubik's Cube Solving Robot

- Built an autonomous robot integrating Python, computer vision, Raspberry Pi/Linux, actuator control, CAD, and 3D-printed components; developed software to capture cube state, compute solutions, and coordinate motion end to end. Managed with Git and structured debugging, documented with a full build guide and BOM.

Full Stack Wordle Game

- Co-developed a browser-based word game in Python, JavaScript, HTML/CSS, SQL, and YAML using Scrum boards and GitHub, iterating on UI/UX from user feedback.

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Education

B.S. Computer Science – Eastern Oregon University (2024) – GPA ~3.6

- Coursework: Software Engineering, Application Security, Operating Systems, Data Structures, C/C++ Programming, Object-Oriented Programming, DBMS (SQL/MySQL)

Certifications

- Ignition Core Certification – Inductive Automation – 2026
- Ignition University Credential – Inductive Automation – 2025
- Responsive Web Design — freeCodeCamp – 2024
- Git Essentials – LinkedIn Learning – 2024
- C# (Microsoft / freeCodeCamp) – freeCodeCamp – 2024